EE 302: Control systems

Quiz-1, Full marks = 20, Time: 1 hour

Date: 30/01/2024

Question 1. Consider the circuit shown with the values of components as specified in the figure 1. [12]

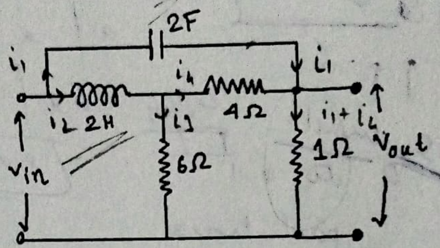

Figure 1

(a) Find the transfer function $\frac{V_{out}(s)}{V_{in}(s)}$.

(b) If a step voltage of 10 V is applied across the terminals of v_{in}, then what would be the final value of current in the inductor. How would one utilize the transfer function in part (a) to obtain the final value of the current in the inductor ?

(c) What is the final value of the voltage across the capacitor, if a step voltage of 10 V is applied across the terminals of v_{in}. Explain how the transfer function of part (a) can help in obtaining this answer ?

(d) What is the pick voltage that the capacitor exhibits when a step input of 10 V is applied as v_{in}?

Question 2. Consider the mechanical system as shown in the figure 2.

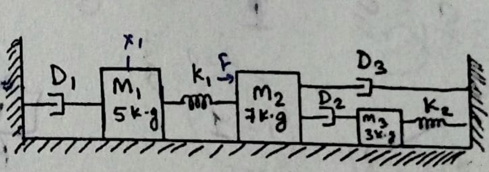

Figure 2

The values of the dampers D_1, D_2 and D_3 are 3, 1 and 3 N-s/m respectively. The values of k_1 and k_2 are 3 and 2 N/m respectively. A force F pushes mass M_2 towards M_3. If $X_1(s)$ is the Laplace transform of position (equilibrium) of mass M_1, then find the transfer function $\frac{X_1(s)}{F(s)}$. [8]

-Paper ends-

-Best of luck-

48
25
73

$4s^2+4s+3s+3$

$4s(s+1)+3(s+1)$

$\frac{-24 \pm \sqrt{(24)^2 - 4(3)(20)}}{40}$

$\frac{-24 \pm \sqrt{336}}{40}$

576